

# JIAYU (RYAN) ZUO

Sophomore Undergraduate  
College of Engineering · EE/BME

 HomePage

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 Shenzhen, China

 Jiayu Zuo



## SELF-INTRODUCTION

Hi! I'm a sophomore undergraduate student in the College of Engineering at Southern University of Science and Technology (SUSTech), intending to major in EE/BME. Strong interests in neuroscience and BCI, with goals to explore the mysteries of the brain and develop innovative brain-computer interface technologies.

Currently conducting research at NCC Lab under the supervision of Associate Professor Quanying Liu. For my latest information, please see my personal homepage.

## SKILLS

**Programming:** Python, C, Java, MATLAB, LaTeX

**Frameworks:** PyTorch, NumPy, MNE

**Tools:** Git, Docker, Jupyter

**Languages:** Chinese (Native), English (Fluent, CET-4)

## RESEARCH PROJECTS (LEADING)

05/2026 – Present

**EEG-Based Visual Decoding in Real-World Natural Scenes**  
• Designed a decoding framework targeting single-trial EEG responses to complex natural scene stimuli, eliminating reliance on trial-averaging that limits real-world applicability.  
• Proposed a subject-agnostic alignment strategy to achieve robust cross-subject generalization without per-subject fine-tuning.  
Single-Trial Decoding / Cross-Subject Generalization / Natural Scene / EEG-to-Image

NCC Lab, SUSTech

03/2026 – Present

**Adolescent Brain Development Assessment via Large-Scale fMRI Foundation Model**  
Core Member  
• Building a foundation model pre-trained on a large-scale private fMRI dataset of Chinese school-age children and adolescents to capture developmental brain patterns.  
• Designing downstream evaluation tasks for cognitive and neurodevelopmental assessment, including age prediction and developmental trajectory modeling.  
fMRI Foundation Model / Brain Development

NCC Lab, SUSTech

09/2025 – Present

**EEG-to-Text Decoding via Foundation Model Alignment**  
• Developed a brain-to-text decoding pipeline aligning EEG representations with large language model embedding spaces via contrastive and generative objectives.  
• Extended decoding to multilingual settings, bridging neural EEG signals and natural language text across languages.  
• Funded by the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds).  
EEG-to-Text / Contrastive Learning

NCC Lab, SUSTech

06/2025 – 09/2025

**Multi-View 3D Reconstruction System for EEG Electrode Localization**  
• Developed a precise electrode positioning system using multi-view 3D reconstruction techniques  
• Implemented source localization algorithms and VGGT for visual reconstruction  
• Applied YOLO-based object detection for automated electrode identification  
• Filed one patent application (under review)  
3D Gaussian Splatting / YOLO / Source Localization

NCC Lab, SUSTech

## EDUCATION

08/2024 - 06/2028

**Bachelor of Engineering**  
Intended Major: EE/BME  
**Completed:** Calculus, Linear Algebra, College Physics, Fundamentals of Electronic Circuits  
**In Progress:** Data Structures and Algorithms, Probability & Statistics, Signals and Systems

Southern University of Science and Technology (SUSTech)

## PUBLICATIONS & UNDER REVIEW

*\*Equal Contribution*

**BHA/DHA** *(title abbreviated to avoid blind review issues)*  
D. Li\*, J. Zuo\*, K. Xie, W. Yang, Y. Liu, C. Wei, & Q. Liu  
Under Review (NeurIPS 2026)

**FlatCLIP** *(title abbreviated to avoid blind review issues)*  
M. Wang, W. Ye, Z. Ning, J. Zuo, J. Xia, H. Wen, & Q. Liu  
Under Review (NeurIPS 2026)

## HONORS & AWARDS

|         |                                                                                                                                                                                                                                            |                                               |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 05/2026 | <b>College Students' Innovative Entrepreneurial Training Plan Program</b>                                                                                                                                                                  | SUSTech                                       |
|         | <ul style="list-style-type: none"><li>Project: <i>Opti-Rhythm: A Digital Sleep Modulation System Based on a Physiological-Topology Brain-Body Coupling Foundation Model</i>.</li><li>Selected with a grant of <b>CNY 20,000</b>.</li></ul> |                                               |
| 12/2025 | <b>Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation</b> ("Climbing Program" Special Funds)                                                                                         | SUSTech                                       |
|         | <ul style="list-style-type: none"><li>Project: <i>Foundation Model-Aligned Chinese Semantic Decoding from Brain Signals</i>.</li><li>Selected as a top-tier undergraduate project (Top 30%) with a grant of <b>CN¥20,000</b>.</li></ul>    |                                               |
| 11/2025 | <b>Award of Excellence (Poster Presentation)</b>                                                                                                                                                                                           | Department of Biomedical Engineering, SUSTech |
|         | <ul style="list-style-type: none"><li>Title: <i>MindPilot: EEG-guided Brain Modulation using Closed-loop Visual Stimulation Optimization</i>.</li><li>Awarded at the 6th SUSTech BME Research Day.</li></ul>                               |                                               |

## COMPETITIONS

|         |                                                                                                                                                                                                                                                                                                                                                                                          |                           |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 05/2026 | <b>Finalist (About Top 1%), 2026 Mathematical Contest in Modeling (MCM)</b>                                                                                                                                                                                                                                                                                                              | COMAP                     |
|         | <ul style="list-style-type: none"><li><b>Team Leader &amp; Advisor</b>. Topic: "<i>Artemis's Choice: Chariot or Elevator?</i>".</li></ul>                                                                                                                                                                                                                                                |                           |
| 12/2025 | <b>Winning Prize, 2025(6th) "Greater Bay Area Cup" Financial Mathematical Modeling Contest</b>                                                                                                                                                                                                                                                                                           | GDSTA                     |
|         | <ul style="list-style-type: none"><li><b>Team Leader</b>. Topic: "<i>Comprehensive Evaluation and Development Analysis of Stablecoins</i>".</li><li>GDSTA: Guangdong Association for Science and Technology</li><li>Hosted by Guangdong Society for Industrial and Applied Mathematics &amp; Guangdong-Hong Kong-Macau Center for Applied Mathematics(Sun Yat-sen University).</li></ul> |                           |
| 05/2025 | <b>Honorable Mention (H Award), 2025 Mathematical Contest in Modeling (MCM)</b>                                                                                                                                                                                                                                                                                                          | COMAP                     |
|         | <ul style="list-style-type: none"><li><b>Solo Participant</b>. Topic: "<i>Cybersecurity Analysis Based on the Global Cybersecurity Index (GCI)</i>".</li></ul>                                                                                                                                                                                                                           |                           |
| 2023    | <b>National Second Prize, China High School Mathematical Contest in Modeling</b>                                                                                                                                                                                                                                                                                                         | Beijing Normal University |
|         | <ul style="list-style-type: none"><li><b>Solo Participant</b>. Topic: "<i>Game Theoretic Analysis of Medical Resource Runs During the COVID-19 Pandemic</i>".</li></ul>                                                                                                                                                                                                                  |                           |

## RESEARCH EXPERIENCE

### Omni-Intelligence

06/2026 – Present

Research Intern working on human preference-aligned post-training for generative models.

Shenzhen  
Supervisors: Dr. Chen Wei & Mr. Dongyang Li

### Neural Computing and Control Lab (NCC Lab)

09/2024 – Present

Undergraduate Researcher focusing on BCI, CHI, and NeuroAI.

Department of Biomedical Engineering, SUSTech  
Supervisor: Associate Professor Quanying Liu